Seunghyun Yoo

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Research Interests

Mechanistic interpretability, diffusion language models, AI safety, and technical AI governance

Education

Purdue University Aug. 2024 – Expected Dec. 2027
B.S. in Artificial Intelligence; Minor in Mathematics *West Lafayette, IN*

Selected Publications and Preprints

- **Seunghyun Yoo**. *White Box Evidence Packages for Policy Audit Reports*. Second Workshop on Technical AI Governance Research (TAIGR), ICML 2026. Sole author.
- Meliksah Yorulmazlar, Justice Sefas, **Seunghyun Yoo**, and Lulu Shao. *Feature Evolution in Diffusion Language Models*. Accepted to the Non-Autoregressive Language Models Workshop, COLM 2026; poster.
- *LLMs as Deceptive Agents: How Role-Based Prompting Induces Semantic Ambiguity*. Preprint, 2025.

Research Experience

Undergraduate Researcher (Team Lead) Jun. 2025 – Present
Governance and Responsible AI Lab (GRAIL) / Purdue AI Forge *Purdue University*

- Designed and implemented a policy audit evaluation spanning 60 AGORA cases, 600 reports, and ten evidence conditions. Demonstrated that audit reports can cite irrelevant internal model evidence despite appearing plausible; authored the resulting TAIGR paper.
- Built the policy representation analysis from corpus preparation through SAE extraction, statistical testing, and causal interventions in Gemma 2 2B. Analyzed 2,438 records and identified 35 layer 24 features stable across five permutation seeds.
- Developed AGORA-RAGTruth, a policy hallucination benchmark with manual span annotations and governance tags, and designed mechanistic analyses separating retrieved context use from parametric knowledge.
- Annotated and validated AGORA policy documents and redesigned the Explore-AGORA graph schema to connect policy passages with risks, strategies, and institutional metadata.

Research Fellow Sep. 2026 – Present
Supervised Program for Alignment Research; Mentors: Yuxiao Li and Di Wu *Remote*

- Studying how latent influence propagates across interacting agents, using activation steering and representation-level analysis to measure behavioral and internal-state shifts over extended conversations.

Research Fellow Feb. 2026 – Jun. 2026
Algoverse AI Safety Fellowship; Supervisor: Justice Sefas *Remote*

- Designed and implemented analyses of SAE features in Dream-v0-Base-7B across six layers and 64 denoising steps, including activation statistics, feature interpretation, and token commitment visualizations.
- Identified recurring patterns of masked token tracking, activation near token commitment, and selective responses to unmasked tokens. Found limited evidence for clean early or late feature stages in the studied setup.
- Analyzed token unmasking across mathematical, natural language, and code reasoning prompts. The resulting paper, *Feature Evolution in Diffusion Language Models*, was accepted to the Non-Autoregressive Language Models Workshop at COLM 2026.

Selected Projects

MARGIN Arena: Agents Under Resource Limits Sep. 2026
AI Incident Response Sprint / Apart Research *Remote*

- Designed and published a Python environment for studying agent behavior under five independently configurable limits on tokens, time, verification calls, tool calls, and sessions.
- Integrated Inspect and ControlArena tasks with original scoring, preserved execution records, and optional activation capture for interpretability. Validated resource accounting, stopping behavior, and analysis interfaces with 50 software tests.

Selected Projects (continued)

Denoising Trajectories as Early-Warning Signals for Diffusion Language Model Control Mar. 2026
AI Control Hackathon; Apart Research / Redwood Research *Remote*

- Built a monitoring and steering system for LLaDA-8B-Instruct using pretrained SAEs, with executable shell benchmarks and tests under prompt and payload variation.
- Detected attacks at mean denoising step 9.60 versus 63.00 for output checks on the main benchmark; early detection persisted under variation (10.53 versus 63.00).

Prompt Injection and Mechanistic Analysis in RAG Feb. 2026
CS 390, Purdue University *West Lafayette, IN*

- Built a BM25 RAG attack evaluation on HotpotQA for Qwen2.5 0.5B and Gemma 2 2B. Combined attention analysis, linear probes, activation patching, and sparse feature analysis to localize attack effects.
- Found the strongest patching recovery in late Qwen layers and sparse late Gemma features activated by attacks; compared attack defenses and their effects on answer quality.

Prompting and Retrieval Evaluation Jan. 2026 – Mar. 2026
CS 390, Purdue University *West Lafayette, IN*

- Evaluated Llama 3.1 8B Instruct on 1,110 HotpotQA examples. Wikipedia retrieval reached 43.65 F1 versus 28.34 for BM25 demonstration retrieval at their best tested settings; compared example selection and context placement.

Teaching, Mentoring, and Outreach

Undergraduate Teaching Assistant, TDM 101/102 Fall 2025
Purdue University *West Lafayette, IN*

- Supported instruction and student problem solving in foundational data science coursework.

Founder & Director Jan. 2025 – Present
Carter Student AI Research *Remote*

- Organized weekly seminars on AI alignment and mechanistic interpretability; mentored peers through reading and research projects.
- Delivered machine learning, deep learning, and NLP lectures to college and high school audiences, including students at USC and Binghamton (Jan. 2025 – Aug. 2025).

AI Study Leader Sep. 2025 – Present
LikeLion US / OZ Lab *West Lafayette, IN*

- Teach weekly sessions on LLM From Scratch, MiniArena, and Interpretability for Purdue's Korean student community.

Additional Experience

Short Course Tester Dec. 2024 – Apr. 2025
DeepLearning.AI *Remote*

- Validated course content and exercises in RAG, Transformers, TensorFlow, and agent tools before release; documented test cases and technical issues through GitHub.

Selected Posters and Presentations

- *Co-CoT: A Prompt-Based Framework for Collaborative Chain-of-Thought Reasoning*. RISE AI Conference, Oct. 2025. Poster / presentation; first author.
- *Factually Grounded and Human-Aligned RAG Systems for AI Governance and Policy*. Purdue URC, 2025. Poster.
- Internal GRAIL presentations: *Explore-AGORA 2* (Feb. 2026) and *DPO + Explore-AGORA* (Nov. 2025).

Honors and Recognition

Purdue PLaCE Student Spotlight, 2024. Travel Grant Recipient, RISE AI Conference, University of Notre Dame.

Technical Skills

Programming and ML Python, PyTorch, Hugging Face Transformers, Git.
Language models RAG, diffusion models, model evaluation.
Interpretability Sparse autoencoders, activation patching, probing, TransformerLens, NNsight, NDIF.

Activities

AI Safety Purdue and ML@Purdue, general member; Purdue eSports, Overwatch Junior Varsity.